Activity 1.1.3 Mechanical System Efficiency

MECHANICAL SYSTEM EFFICIENCY

Investigating Energy Conversions

GOALS | MATERIALS | RESOURCES

- Build and test an energy conversion system to define and calculate efficiency, work, and power.

GOALS | MATERIALS | RESOURCES

- VEX® V5 POE Kit
- Materials and equipment listed in the POE Gearbox Motor Winch Construction Guide
- Permanent marker

- Timing device to measure time to 0.01-second precision

GOALS	MATERIALS	RESOURCES

- **POE Gearbox Motor Winch Construction Guide**
- **POE Winch Mass Construction Guide**
- **Multimeter Operation**
- **Breadboards**
- **Activity 1.1.3 Mechanical System Efficiency (Downloadable PDF)**

Energy cannot be created nor destroyed, but it can be converted from one form to another. Engineers design **energy conversion** systems to convert input energy into a different form of output energy. Within any conversion system, some input energy is changed into less desirable forms of energy such as **geothermal energy** because of **resistance** and **friction** in the conversion process. To minimize undesirable energy conversions within a system, known as energy "losses," engineers optimize system **efficiency**.

In this activity, you will investigate an energy conversion system designed to change electrical energy into mechanical energy. You will then determine the efficiency of the system.

Introduction to Energy, Work, and Power

Energy, work, and power are terms you are likely familiar with and often use in your daily lives. You may hear someone explain that they don’t quite have enough energy to go for a run after school because they stayed up too late the night before. You may have heard someone discuss how much power their pickup truck has. But what are these terms and what do they really mean when we approach them from a physics perspective?

Energy

A famous idea that you may be familiar with is, "Energy cannot be created nor destroyed." The statement took hundreds of years to confirm through the effort and collaboration of many brilliant scientific minds. Though it can neither be created nor destroyed, energy can be transferred from one form to another. This concept is known as the First Law of Thermodynamics or the Law of Conservation of Energy.

Energy can exist in two states: **potential** and **kinetic**. It can also transfer from one form to another. Explore Potential Energy, Kinetic Energy, and Energy Transfer to learn more about the two states of energy and the process of moving from one form to another.

POTENTIAL ENERGY	KINETIC ENERGY	ENERGY TRANSFER

Potential energy manifests in various forms:

- **Gravitational potential energy**. Lifting an object away from the earth’s surface gives the object gravitational potential energy because the object has moved against the earth's gravity. A helicopter has gravitational potential energy.

- **Elastic energy**. Stretching a rubber band gives the rubber band electromagnetic potential energy because the rubber band's molecules have been forced apart, overcoming the molecules' electromagnetic attraction.
- **Electrical energy**. Applying a voltage to a conductor gives the conductor electromagnetic energy because the electrons have not yet yielded to the electromagnetic force.
- **Chemical energy**. The electrons in molecules of fuel have electromagnetic potential energy. They are too far away from the nuclei of the product that can be formed during combustion.
- **Thermal or heat energy**. Changes among solid, liquid, and gaseous states are electromagnetic potential energy because the molecules have overcome the attraction between electrons in one molecule and the nuclei of neighboring molecules.

POTENTIAL ENERGY | KINETIC ENERGY | ENERGY TRANSFER

Kinetic energy manifests in various forms:

- **Energy of motion**. The movement of whole objects; this is usually the type of motion meant when discussing kinetic energy. Windmills use the kinetic energy of motion to generate power.
- **Electrical energy**. The movement of electrons in a conductor as they respond to an applied voltage. The electrical current that carries a charge from a power source to a light bulb is kinetic energy.
- **Thermal or heat energy**. The movement of molecules and electrons. A space shuttle uses kinetic energy to enter and maintain orbit by using its thrusters and engines to continuously increase speed and acceleration.

POTENTIAL ENERGY | KINETIC ENERGY | ENERGY TRANSFER

Energy transfer involves the conversion of one form of energy into another form. Examples of energy transfer include:

- **Chemical to Kinetic**. Food is converted, through several biological processes, into motion for playing sports.
- **Radiant into Chemical**. Light from the Sun is transferred by plants into chemical energy that it uses to grow.
- **Electrical to Thermal**. Energy transferred to an oven is converted to thermal energy for heating food.

Efficiency

Efficiency (symbol η) is the percentage of input energy (E_{in}) that is transformed to output energy in the desired form (E_{out}).

$$\eta = \frac{\mathrm{E_{out}}}{\mathrm{E_{in}}}$$

When energy is converted from one type to another, the goal is to maximize the efficiency of that process. For example, most electric power plants have an efficiency of around 30%. The rest of the energy is often lost through the smokestack as thermal energy. However, some power plants follow a process called **cogenerating**, where the system captures this thermal energy and uses it to heat a building, for example. At that point, the efficiency can measure as high as 80%. Efficiency is usually written as a percentage which means taking the same efficiency equation and multiplying it by 100%. For example, if the efficiency for a system is measured at 0.75, you can multiply by 100% to get 75% efficiency.

Work

Work (symbol W) is the energy transferred when a force, F, is applied to an object moving through a distance, d.

$$W = F \cdot d$$

You complete work on a regular basis, but no, reading and exploring the world of energy in Principles of Engineering is not considered work! But if you have ever climbed up stairs, pushed a table or desk across a room, or lifted a book off of the ground, then you have done work. Work is done only by the component of force parallel to the distance the object moves, $F_{||}$. Work is measured in Joules (J) and 1 J = 1 N*m.

Work Example Problem

A student lifts a 50 pound ball 4 feet. How many joules of work did the student do?

Step 2

Work Formula

Substitute the known values in the work formula.

$$\mathrm{W} = \mathrm{F} \cdot \mathrm{d}$$

$$\mathrm{W} = 50\,\mathrm{lb} \cdot 4\,\mathrm{ft}$$

$$\mathrm{W} = 200\,\mathrm{ft} \cdot \mathrm{lb}$$

The answer 200 ft-lb is an amount of work, but it is not in the unit of Joules. Therefore, it needs to be converted.

Step 3

Conversion to Joules

Convert 50 pounds to Newtons using the constant of 1 N for every 0.225 pound. Then convert 4 feet into meters (m) using the constant of 3.28 feet for every 1 meter.

$$50\,\mathrm{lb} \cdot \frac{1\,\mathrm{N}}{0.225\,\mathrm{lb}} = 222.5\,\mathrm{N}$$

$$4\,\mathrm{ft} \cdot \frac{1\,\mathrm{m}}{3.28\,\mathrm{ft}} = 1.22\,\mathrm{m}$$

Step 4

Solve for Work

Substitute the new known values in the work formula.

$$\mathrm{W} = \mathrm{F} \cdot \mathrm{d}$$

$$\mathrm{W} = 222.5\,\mathrm{N} \cdot 1.22\,\mathrm{m}$$

$$\mathrm{W} = 271\,\mathrm{J}$$

Power

Power (symbol P) is the rate at which work (W) is performed in a given time (t).

$$\mathrm{P} = \frac{\mathrm{W}}{\mathrm{t}}$$

Power is found in various forms:

- Electrical power uses electrical energy to do work.
- Mechanical power uses mechanical energy to do work.
- Fluid power uses energy transferred by liquids (hydraulic) and gases (pneumatic) to do work.

The previous example problem showed the amount of work done when lifting a 50-pound ball a distance of 4 feet. But what if time was also measured?

Power Example Problem

A student lifts a 50 pound ball 4 feet in 5 seconds. How much power was used to lift the ball?

Step 2

Power Formula

Substitute the known values in the power formula. Recall, W = 271 J, as determined in the previous example problem.

$$\mathrm{P} = \frac{\mathrm{W}}{\mathrm{t}} = \frac{271\,\mathrm{J}}{5\,\mathrm{s}} = 54.3\frac{\mathrm{J}}{\mathrm{s}}$$

Step 3

SI Units of Power

This value can also be expressed in the SI units for power, which is Watts.

$$\frac{\mathrm{J}}{\mathrm{s}} = \mathrm{W}\text{, therefore...}$$

$$54.3\ \mathrm{W}$$

Electrical power is measured by multiplying the current (I) by the voltage (V) of the system. You will examine electrical circuits more closely in Unit 3 Energy In Action.

P = I·V

Data Collection

1 To get acquainted with the tools you will use to complete this activity, study the **Breadboards** and **Multimeter Operation** resources.

2 Build the winch and mass using the **POE Gearbox Motor Winch Construction Guide** and the **POE Winch Mass Construction Guide**. The completed assembly is shown in Figure 1.

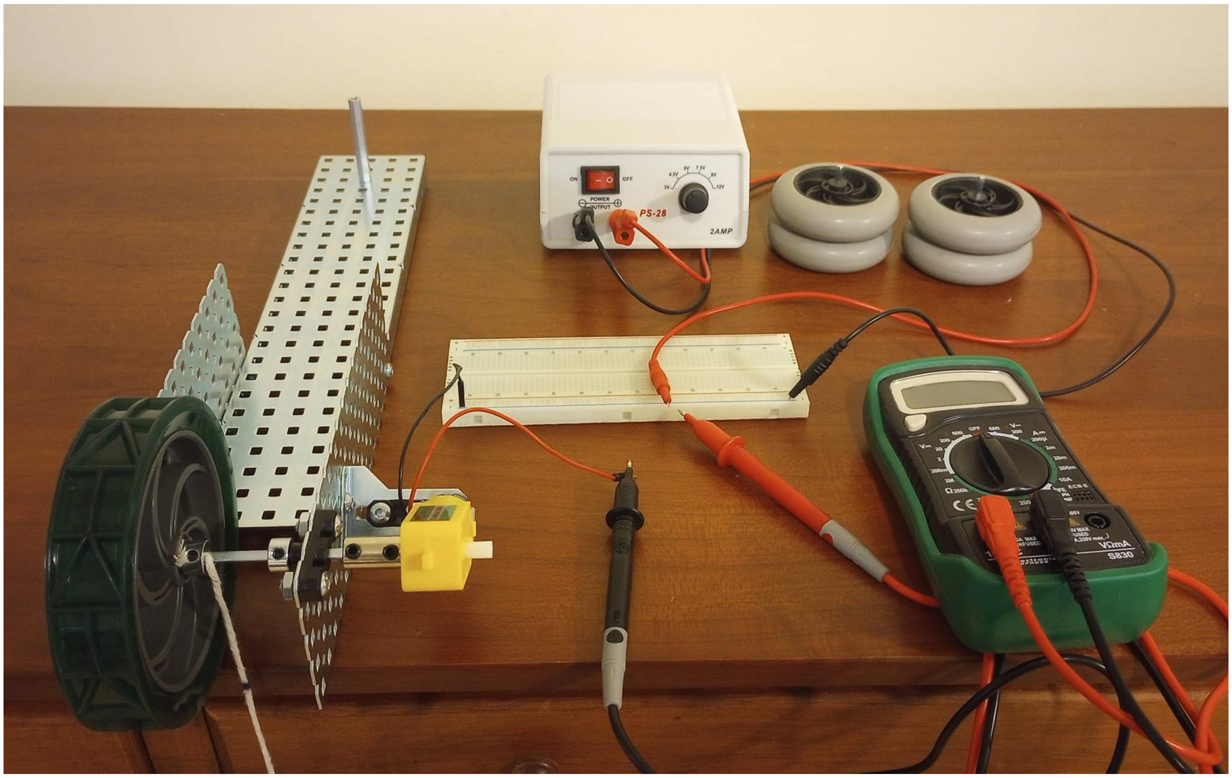

Figure 1. POE Gearbox Motor Winch Assembly

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Attach Mass 1 to the end of the winch.

- Insert the loop at the end of the cable through the spokes of both wheels of a single mass as shown in Figure 2.
- Hitch the loop over the end of the shaft as shown in Figure 3.
- Place the winch on the edge of a table so that the mass freely hangs from the edge as shown in Figure 4.

NOTE: The winch was designed to provide a counterbalance (using the long C-channel) to the downward force of the mass on the winch.

Figure 2. Winch Cable Through Wheel Spokes

Figure 3. Winch Mass 1 Attached to Winch Cable

Figure 4. Winch Mass 1 Hanging from the Winch Cable

4 Using a permanent marker, mark the winch cable around its entire circumference at approximately 5 cm (2 inches) below where the cable wraps around the shaft as shown in Figure 5.

5 Measure 30.0 cm (approximately 11-3/4 inches) down from the first mark, and mark the winch cable around its entire circumference at a second point as shown in Figure 5. These are your winch start and stop points.

Figure 5. Start and Stop Indicator Marks

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Measure the length to be lifted, which is the distance between the start and stop points in meters.

NOTE: Units and precision are critical to record as part of the data in your PLTW Engineering Notebook.

Additional Information

- The Winch Mass Construction Guide produces two assemblies, each with a mass of 100.0 grams. Assume the acceleration due to gravity is 9.81 m/s^2.
- Also, throughout this activity, you will see the terms **current** and **voltage**. These ideas will be explained in more detail in Lesson 3.1 Electrical Circuits.

Safety Reminder: Since this is a combination of low voltage and low current, there is no risk of electrocution. However, this is not a typical way to make an electrical connection. Never attempt to hold live wires together outside of this controlled lab environment.

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Follow the remaining steps for the winch to lift the mass 30.0 cm.

- Turn on the power supply.
- Turn the voltage knob to 4.5 V as shown in Figure 6.
- Turn the multimeter rotary dial to 10 A as shown in Figure 7.
- Touch and hold the red lead from the power supply to the red multimeter test probe as shown in Figure 8.
- Touch and hold the black multimeter test probe to the red lead from the motor as shown in Figure 8.

NOTE: The motor will rotate when you make this connection, as this will complete the circuit! To stop rotation, simply stop holding two of the wires together.

- Start the stopwatch when the start mark reaches the shaft.

- Record the power supply voltage along with the current displayed on the multimeter.
- Stop the stopwatch when the stop mark reaches the winch shaft.
- Record the following parameters:
  - Mass lifted (converted from grams to Newtons)
  - Voltage
  - Current
  - Time

NOTE: You can guide the winch cable with your finger so the cable wraps neatly on the winch drum as shown in Figure 9.

Figure 6. Power Supply

Figure 7. Multimeter Setting

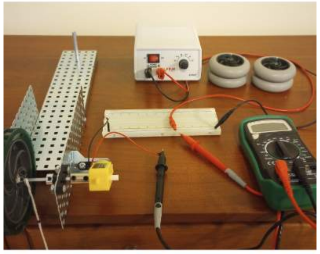

Figure 8. Completing the Circuit

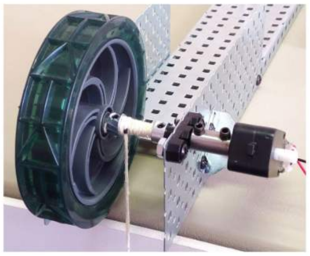

Figure 9. Winch Drum with Uniform Windings

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Using one of the following methods, unwind the winch cable so that the start mark on the winch cable is lined up with a starting point:

- Turn off the power supply.
- Turn the winch wheel by hand until the winch cable is fully unwound.
- Turn on the power supply.

9 Add Mass 2 using the following steps:

- Unhitch the loop from the shaft of Mass 1.
- Pass the cable loop through the spokes of the two wheels in Mass 2.
- Hitch the loop over the end of the shaft as shown in Figure 10.

Figure 10. Mass 1 and Mass 2

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10 Repeat steps 7 and 8 using two mass assemblies. Record the following parameters for the two mass assemblies:

- Mass to be lifted (convert to N).
- Voltage

- Current
- Time

Data Analysis

Work involves the amount of force (F) exerted over a specific distance (d). Work is not related to time. If two winches lift identical mass assemblies the same distance, they do the same work, even if one winch takes longer. Use the following formula to determine how many joules (J) of work it took to lift the mass in the system. As you carry your units through to the solution, change the final answer from Newton-meters (N·m) to the equivalent in joules (J). You can do this through the conversion ratio of 1:1 meaning that 1 N·m = 1 J.

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Using the formula for work, calculate the work done by the winch system in units of Newton-meters.

NOTE: **Power** is a function of time, force, and distance. Two winch operations—one slow and one fast—do the same amount of work to lift the same mass because both apply the same force and move a mass over the same distance. Time is not a factor, but to cover the distance faster, the faster winch requires more power.

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Using the formula for power, calculate the output power of the system in units of watts.

$$P = \frac{W}{t}$$

13 Using the formula for electrical power, calculate the power of the electrical system in units of watts.

$$P = I \cdot V$$

NOTE: Carry your units through to the solution and change the final answer from I·V to the equivalent unit W.

14 Using the formula for efficiency, calculate the efficiency of each system to compare the energy input to the energy output.

$$\eta = \frac{E_{out}}{E_{in}}$$

The energy we are measuring is power, so the equation for efficiency looks like this:

$$\eta = \frac{P_{out}}{P_{in}}$$

Conclusion

Question 1

Describe three factors that reduced efficiency in the winch system.

Question 2

Describe one strategy for making the system even more efficient.

Question 3

Explain at least two more reasons why automotive engineers work to eliminate inefficiency in vehicle designs.